

Top 5 Global Robotics Trends 2026

International Federation of Robotics reports

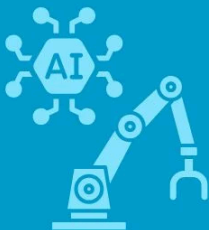
Frankfurt, Jan 08, 2026 — The global market value of industrial robot installations has reached an all-time high of US\$ 16.7 billion. Future demand will be driven by a number of technological innovations, market forces and new fields of business. The International Federation of Robotics reports on the top 5 trends for the robotics industry for 2026.

TOP 5 GLOBAL ROBOTICS TRENDS IN 2026



1

AI & AUTONOMY IN ROBOTICS



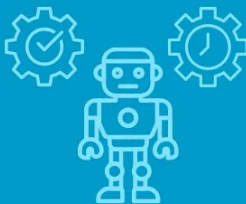
2

ROBOTS GAIN VERSATILITY AS IT MEETS OT



3

HUMANOIDS TO PROVE RELIABILITY AND EFFICIENCY



4

SAFETY AND SECURITY IN ROBOTICS



5

ROBOTS AS ALLIES IN TACKLING LABOR GAPS





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<https://ifr.org/ifr-press-releases/top-5-global-robotics-trends-2026>

Source: International Federation of Robotics

1 – AI & Autonomy in Robotics

Robots that use artificial intelligence to work independently are becoming more common. Main benefit of AI in this context is the increased autonomy of robots empowered by AI. Different types of AI drive this trend: Analytical AI helps to process large datasets, detect patterns, and provides actionable insights. This enables them to autonomously anticipate failures before they occur in smart factories or path planning and resource allocation in logistics for example.

Generative AI, on the other hand, marks a shift from rule-based automation to intelligent, self-evolving systems. GenAI creates new outputs and enables robots to learn new tasks autonomously and generate training data through simulation. This also allows a new kind of human-robot interaction with natural language and vision-based commands.

A key trend to further develop autonomy in robotics is Agentic AI. This technology combines analytical AI for structured decision-making, and generative AI for adaptability. The hybrid approach aims to make modern robotics capable of working independently in complex, real-world environments.

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2 – Robots gain versatility as IT meets OT

Demand for versatile robots is accelerating. This directly reflects a market push toward a convergence of Information Technology (IT) and Operational Technology (OT). The merge of IT's data-processing power and OT's physical control capabilities enhance robotics versatility through real-time data exchange, automation, and advanced analytics. This integration is a foundational element of the digital enterprise and Industry 4.0. The IT/OT convergence breaks down these silos, creating a seamless flow of data between the digital and physical worlds, which significantly enhances the capabilities and versatility of robotics.

3 – Humanoids to prove reliability and efficiency

The field of humanoid robotics is expanding rapidly. Humanoid robots for industrial use are seen as a promising technology where flexibility is required, typically in environments designed for humans. Pioneered by the automotive industry, applications in warehousing and manufacturing are coming into focus worldwide.

Today, companies and researchers are moving beyond prototypes to deploy humanoids in real life. Reliability and efficiency are key to success: In competing with traditional automation, humanoid robots need to match high industrial requirements towards cycle times, energy consumption and maintenance costs. Industry standards also define safety levels, durability criteria and consistent performance of humanoid robots needed on the factory floor. Humanoids intended to fill labor gaps need to achieve human-level dexterity and productivity, key measures to prove real world efficiency.

4 – Safety and Security in Robotics

As robots increasingly operate alongside humans in factories and service settings, ensuring they operate safely is not just important, it's essential for the robotics industry. The AI-driven autonomy fundamentally changes the safety landscape, which makes testing, validation, and human oversight much more complex—but also more necessary. This becomes particularly clear in the intended use of humanoid robots. Robotic systems need to be designed and certified in line with ISO safety standards and clearly defined liability frameworks.

In the context of AI in robotics and the convergence of IT and OT, a spectrum of safety and security concerns arise that demand robust governance and clear assignment of liability. The rapid expansion of robotics systems into cloud-connected and AI-driven environments is exposing industrial production to a growing array of cybersecurity threats. Experts cite a rise in hacking attempts targeting robot controllers and cloud platforms, enabling unauthorized access and potential system manipulation. As robots become more integrated into workplaces, concerns are mounting over the sensitive data they collect — including video, audio, and sensor streams. Deep learning models which are often described as 'black boxes' can produce results that are difficult or impossible to explain, even to their own developers. The legal and ethical ambiguity surrounding liability has prompted calls for clear frameworks to govern AI deployment.

5 – Robots as allies in tackling labor gaps

Employers around the world are struggling to find people with the specialized skills required. These unfilled jobs leave existing staff covering extra shifts, with rising stress and fatigue across all sectors. A key strategy for addressing this issue is to adopt robotics and automation. In this transformation process, employers benefit from taking their human workforce on board. The close cooperation with employees in implementing robots plays a crucial role to ensure acceptance – both in industrial manufacturing settings as well as in the manifold service applications. The benefits that robots deliver, such as tackling labor shortages, taking away routine tasks or opening up new career opportunities, mean that they will be accepted as allies in the workplace. At the same time, robots are a way to make a workplace much more attractive to young people. Companies and governments are pushing skilling and upskilling programs to help workers keeping up with changing skills demand and competing in an automation-driven economy.

Jane Heffner, IFR President presents the Trends at Automate show 2026

TOP 5 Global Robotics Trends 2026 - by Jane Heff

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Contact

International Federation of Robotics

PRESS OFFICER

Carsten Heer

phone +49 (0) 40 822 44 284

E-Mail: press@ifr.org

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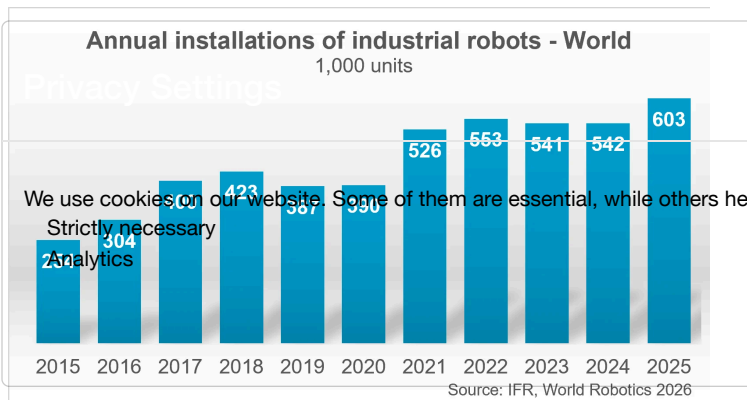
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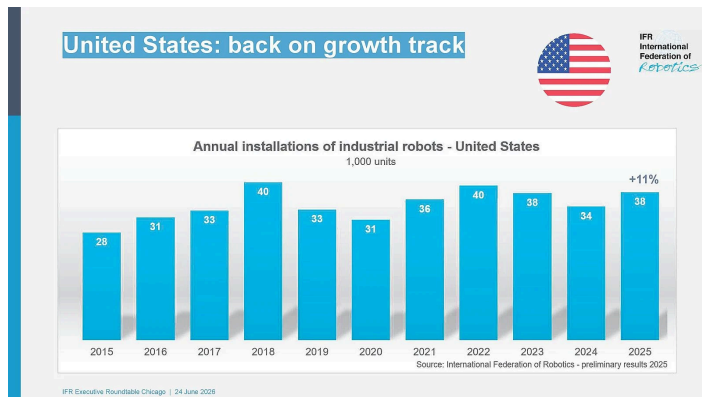
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US Robot Industry Returns to Double Digit Growth (/ifr-press-releases/news/us-robot-industry-returns-to-double-digit-growth)



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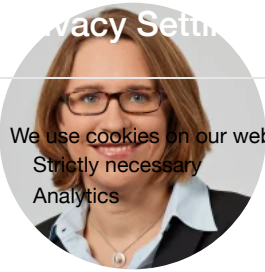
Carsten Heer

IFR Press Inquiries

Phone: +49 40-822 44 284

E-Mail: [press\(at\)ifr.org](mailto:press(at)ifr.org)

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